

RICE LAKE, WI, USA

WMO: 726467

Lat: 45.419N

Lon: 91.773W

Elev: 1105

StdP: 14.12

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-9.6	-21.9	1.7	-13.5	-17.3	2.2	-8.4	20.0	21.3	18.2	18.7	2.3	290	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	87.7	72.7	84.0	70.7	81.5	68.7	75.6	83.1	73.5	80.7	71.6	78.8	10.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	127.9	79.6	71.8	122.3	78.6	69.7	113.6	76.5	39.8	83.3	37.9	80.4	36.1	77.9	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.5	15.8	DB	-23.1	91.8	5.7	3.5	-27.2	94.4	-30.6	96.4	-33.8	98.4	-37.9	101.0	(4)
(5)			WB	-23.4	79.1	5.7	3.5	-27.5	81.6	-30.8	83.6	-34.0	85.6	-38.1	88.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.7	14.4	17.2	30.1	44.1	55.5	65.1	69.7	66.8	59.8	46.6	33.6	20.0		
	DBStd	21.60	12.88	12.73	12.34	9.62	8.29	6.48	5.49	5.55	8.16	8.70	9.90	12.21		
	HDD50	4524	1104	917	626	223	38	0	0	0	14	172	499	931		
	HDD65	8177	1569	1337	1082	628	313	77	19	45	196	572	943	1396		
	CDD50	2227	0	0	9	47	208	453	609	520	308	67	6	0		
	CDD65	405	0	0	0	1	18	80	164	99	40	3	0	0		
	CDH74	3674	0	0	2	27	227	694	1507	826	363	28	0	0		
	CDH80	923	0	0	0	5	55	153	429	189	90	2	0	0		
(14) Wind	WSAvg	6.3	6.0	6.3	6.6	7.8	7.5	6.2	5.1	4.8	5.9	6.6	6.6	6.1		
(15) Precipitation	PrecAvg	33.0	1.1	0.9	1.9	3.0	3.6	4.6	4.0	4.4	3.5	2.9	1.9	1.1		
	PrecMax	42.6	2.7	2.1	4.6	6.4	6.4	7.7	7.6	9.1	7.4	6.6	6.7	2.2		
	PrecMin	23.7	0.2	0.2	0.7	0.7	1.4	1.7	1.7	0.7	0.8	0.8	0.1	0.3		
	PrecStd	4.7	0.6	0.5	1.0	1.4	1.4	1.7	1.5	2.0	1.7	1.6	1.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	48.1	68.2	78.5	86.5	88.4	91.2	90.5	87.9	78.7	64.4	48.2		
		MCWB	38.0	43.0	56.3	60.5	66.7	73.0	75.1	73.7	71.3	62.5	54.3	45.9		
	2%	DB	37.2	43.2	59.4	71.5	80.6	84.1	88.0	84.3	82.0	72.1	58.5	42.3		
		MCWB	34.3	39.2	50.9	55.1	63.1	69.3	74.2	71.8	68.6	60.8	51.3	38.8		
	5%	DB	34.4	38.7	52.9	65.9	75.2	81.2	84.2	81.5	78.6	66.2	53.5	38.4		
		MCWB	32.9	35.7	47.0	51.5	60.4	67.6	71.8	69.3	66.5	57.4	47.0	36.1		
	10%	DB	32.1	35.5	46.0	61.2	71.2	78.6	81.7	79.0	74.0	61.5	47.9	35.7		
		MCWB	30.9	33.2	41.0	48.8	58.2	65.7	69.9	67.8	64.2	53.5	43.3	33.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	43.6	59.2	62.8	71.7	76.5	79.9	77.0	74.3	69.6	57.6	45.7		
		MCDB	41.9	46.1	65.6	73.9	81.8	84.8	86.4	84.1	82.1	71.9	61.4	46.6		
	2%	WB	35.2	39.5	52.9	57.2	66.4	73.8	76.5	74.3	71.3	62.9	52.7	39.0		
		MCDB	37.3	42.8	57.1	68.1	75.2	80.4	84.1	81.5	79.1	67.8	58.0	40.7		
	5%	WB	33.2	36.0	46.5	53.2	63.3	70.8	74.2	72.1	68.6	58.5	47.0	36.2		
		MCDB	34.4	38.5	51.8	62.3	70.7	77.2	81.4	79.0	75.7	65.1	51.4	38.1		
	10%	WB	30.5	32.7	41.3	50.2	60.7	68.1	72.2	69.9	66.1	54.6	43.5	33.6		
		MCDB	31.8	34.7	46.1	59.8	68.0	75.0	79.5	76.2	72.4	60.4	47.9	35.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	18.7	18.2	20.2	21.0	19.8	20.7	20.4	21.1	18.1	14.8	14.3		
		MCDBR	15.8	17.3	23.5	29.1	27.1	23.0	23.2	23.1	25.1	26.0	23.6	16.2		
	5% WB	MCWBR	13.7	13.6	14.7	15.5	12.7	11.1	11.6	12.4	13.9	15.5	15.9	13.4		
		MCWBR	14.9	15.1	21.9	25.1	21.5	19.2	20.5	20.6	21.4	20.9	21.0	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.272	0.310	0.351	0.384	0.385	0.374	0.385	0.359	0.315	0.285	0.261		
	taud		2.385	2.329	2.360	2.364	2.311	2.339	2.389	2.372	2.423	2.532	2.514	2.428		
	Ebn at Noon		277	291	293	288	280	280	281	274	272	273	264	262		
	Edh at Noon		23	30	33	36	40	39	36	36	31	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		526	877	1232	1453	1687	1863	1948	1631	1270	799	490	362		
	RadStd		57	75	113	147	130	145	77	115	83	95	48	43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-2.29	N/A	-1.94	-2.36	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page